

Pulkit Garg

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EDUCATION

University of Waterloo — Bachelor of Computer Science

Sep 2022 – Present

Minor in Cognitive Science · Fine Arts Specialization · President's Scholarship of Distinction

TECHNICAL SKILLS

Languages: C++, C#, Python, Java, Kotlin, GLSL, HLSL

Graphics & Systems: Vulkan, OpenGL, GPU Programming, Compute Shaders, Multithreading, Real-Time Rendering

Tools & Frameworks: Git, CI/CD, Unit & Integration Testing, Unity, Unreal Engine, React Native, Firebase

EXPERIENCE

Graphics Programming Intern — SideFX

Jan 2026 – Present

- Implemented core 3D viewport functionality in Vulkan and built multiple Vulkan libraries with custom GLSL shaders, achieving >20× polygon reduction across instances and significant draw-call reduction.
- Designed and built a GPU-based Vulkan text-rendering pipeline from scratch using bitmap fonts and texture atlases; implemented skyline bin-packing and extended the system to support TrueType font rasterization.
- Worked directly on performance-critical, cross-platform rendering code used by professional VFX artists in Houdini.

3D Software Development R&D Intern — SideFX

May 2025 – Sep 2025

- Improved data import systems by enabling high-precision numerical parsing and processing for scientific and simulation workflows.
- Reworked mesh decimation algorithms to achieve >10× polygon reduction while preserving topological detail and UV/texture maps.
- Designed and implemented a real-time geometry editing tool based on the Push-Pull algorithm, using custom spatial acceleration structures to reduce user interaction complexity by up to 12× and increase modeling throughput by 30%.

Virtual Reality Developer — Liftwerx

Sep 2024 – Dec 2024

- Developed VR training modules for wind turbine technicians using Unity, Unreal Engine, and Blender.
- Led end-to-end development including scripting, debugging, and deployment on AWS S3 and Meta Quest headsets using a custom Android launcher.
- Created a custom Android plugin enabling centralized data synchronization and real-time reporting on technician performance.

Software Developer Intern — OS & Graphics — Ford Motor Company of Canada

Jan 2024 – Apr 2024

- Contributed to Android Automotive OS by integrating critical vehicle systems including lighting and climate using C++ and Kotlin.
- Reduced memory leaks by refactoring static companion objects, improving OS stability.
- Reduced service load times by up to 70% by implementing multithreaded services.

Software Developer Intern — 3D Visualization — Ford Motor Company of Canada

May 2023 – Sep 2023

- Built real-time 3D visualization tools for next-generation Ford vehicles using Google Filament; implemented asynchronous asset loading to improve load times, graphical fidelity, and user experience.

PROJECTS

FluidSim — C++, Vulkan, GLSL

GitHub

Highly optimized real-time GPU particle-based fluid simulation with photorealistic rendering, running on a custom-built Vulkan graphics engine.

Voksel — C#, Unity, HLSL

GitHub

Cross-platform open-source 3D modeling tool for voxel-based game asset creation.

Pashu — Unity, C#

Real-time strategy game about running an animal shelter, featuring animal AI, resource management, and a time-of-day system; featured in the GitHub Education program portfolio.

SpaceLine — Unity, SteamVR

Physics-based space sandbox with real-time gravity and climate simulation, plus a VR mode built with SteamVR; won NASA Space Apps India and was selected as a global finalist.

AWARDS

Hawk Hacks 2023 — Laurier University · Hack Western 2022 — Best Gaming Hack

NASA SpaceApps 2019 — National Winner & Global Finalist among 30,000 participants · AngelHack 2019 —

AWS Award